MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 11746258
COMPUTER-2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirement for one type of circuit card assembly. The requirements of this specification together with the general requirements of specification MIL-C-48411 comprise the complete requirements for Computer-2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-48411	Circuit Card Assemblies, General Specification for

DRAWINGS

11746258	Circuit Card Assembly, Computer-2

(Copies of specifications, standards, drawings, and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item Definition.- This assembly is the second of six (6) single ended printed circuit boards consisting of operational amplifiers, FET switches, comparator, multivibrator and nand gates necessary to provide voltage to frequency conversion for storage in digital to analog converters.

3.1.1 Major component list.- The components of this assembly shall be as specified on the Engineering Parts List (EPL) 11746258.

3.2 Characteristics.

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3.2.1 Performance.- The performance requirements specified herein shall be met on every assembly at standard ambient conditions prior to Government acceptance.

3.2.1.1 Power supply.- The requirements of 3.2 shall be met with the following power supply voltages.

a. +5 VDC
b. +15 VDC
c. -15 VDC

3.2.1.2 Operational amplifier.- The operational amplifier network shall be capable of integrating its inputs.

3.2.1.3 Multivibrator.- The multivibrator shall be capable of a monostable output.

3.2.1.4 Differential comparator.- The comparator shall be capable of inputs between ± 10 VDC and yeilding a logical output.

3.2,1.5 Integrator.- The network shall be capable of integrating the inputs applied to Pin 24 and yeilding an output on pin 28.

3.2.2 Environmental.- The assembly shall meet the environmental requirements of specification MIL-C-48411.

3.2.3 Physical Characteristics.- The physical characteristics shall be as specified on drawing 11746258 and the engineering parts list.

3.3 Design and Construction.

3.3.1 Production Drawings.- This assembly shall be fabricated in accordance with the requirements of specification MIL-C-48411, Drawing 11746258 and associated lists.

3.3.2 Standard of Manufacture.- The standard of manufacture shall be those specified in specification MIL-C-48411.

3.3.3 Workmanship.- The workmanship shall be as specified in specification MIL-C-48411.

3.3.4 Preproduction sample.- Preproduction sampling shall be as specified in specification MIL-C-48411.

4. QUALITY ASSURANCE PROVISOIONS

4.1 General.

4.1.1 Responsibility for inspection.- The responsibility for inspection shall be as specified in specification MIL-C-48411.

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4.1.2 Special test and examination.- The assembly shall undergo first article and control testing in accordance with 4.1.2 of specification MIL-C-48411.

4.2 Quality conformance inspection.- The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-48411.

4.2.1 Test equipment. (see Figure 1)

4.2.1.1 Power Supply (PS1).- The power supply shall be 10 ± 0.5 VDC, 100 mA or greater and a ripple of less than 10 mV RMS.

4.2.1.2 Power Supply (PS_2 & PS3).- The power supply shall be 20 ± 0.5 VDC, 200 mA or greater and a ripple of less than 20 mV RMS.

4.2.1.3 Oscilloscope.- Rise time of scope and associated measuring probe of 50 nanoseconds of less, sweep period of 50 nanoseconds per centimeter or greater, input resistance greater than 8 megohms, input capacitance less than 25 picafarads.

4.2.1.4 Digital Voltmeter.- The digital voltmeter shall have a voltage accuracy and resolution of 0.1 percent or better from zero to 100 VDC. Input inpedance of 10 megohms or greater on all voltage scales.

4.2.1.5 Dummy loads.- The dummy loads shall be one 3-input positive "Nand" gate or the equivalent to SN54L10, and resistance of 10.000K ohms ± 0.10% 1/2 watt.

4.2.2 Performance.- Unless otherwise specified, the performance tests specified herein shall be satisfactorily met at standard ambient conditions. Standard ambient conditions for the purpose of these tests shall be defined as:

a. Temperature between 60°F and 90°F.
b. Barometric pressure between 28 and 32 inches of mercury.
c. Relative Humidity (RH) up to 90%.

4.2.2.1 Power supply.- All the requirements shall be met with the following power supply voltages.

a. +5 ± 0.05 VDC
b. +15 ± 0.05 VDC
c. -15 ± 0.05 VDC

4.2.2.2 Digital Output.- The "0" or false output level shall be between 0.00 and 0.40 VDC. The "1" or true output level shall be between +2.40 and +5.00 VDC.

4.2.2.3 Test Set-up.- The test of the circuit card assembly shall be performed with the inputs and outputs as diagrammed in Figure 1 unless otherwise specified.

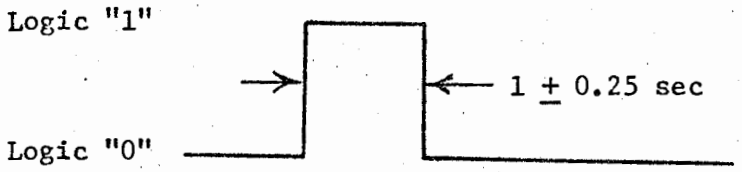
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4.2.2.4 Operational amplifier.- The amplifier network shall have the outputs as shown below for the corresponding inputs.

TABLE I

	Input (VDC ± 1 mV)		Output (VDC ± 0.10 VDC)		
Pin	10	19	4	16	5
	-5.000	0.000	+10.00	5.12	5.71
	2.500	-2.000	0.00	0.00	0.00
	0.000	+2.000	-5.00	2.56	2.86
	5.000	0.000	-10.00	-5.12	-5.71

4.2.2.5 Multivibrator.- When ground is applied to pin 12 the output at pin 1 shall be a pulse of 1 ± 0.25 sec duration as shown below.



4.2.2.6 Differential Comparator.- Apply Logic "1" to pin 11 and 12, 0 VDC to pin 3 and a variable ± 10 VDC to pin 9. When the variable voltage is increased positively from -10 VDC, the output on pin 20 shall switch from Logic "1" to Logic "0" when the voltage exceed 0.055 ± 0.001 VDC. With the variable voltage on pin 9 changed +10 VDC and decreased, the output on pin 18 shall switch from Logic "1" to Logic "0" when the voltage is less than -0.055 ± 0.001 VDC. By applying 0 VDC to pin 9, -10 VDC to pin 3, and Logic "0" to pin 12, the output on pin 20 shall be Logic "1". Apply +10 VDC to pin 3, Logic "0" to pin 12, Logic "1" to pin 11 and the output on pin 18 shall be Logic "1". Apply Logic "1" to pin 12 and Logic "0" to pin 11 and the output on pin 18 shall be Logic "1".

4.2.2.7 Intergrator.- 1. The integrator network shall have the outputs as shown in Table II for the corresponding inputs when pin 12 is at Logic "0", 0 V ± 1 mV is applied to pin 24 and R4 has been adjusted read 0V ± 1 mV on pin 29.

TABLE II

	Input (VDC ± 1 mV)		Output (VDC ± 0.1 VDC)
Pins	24	29	28
	+10.000	-10.000	10.0
	+5.000	-5.000	5.0
	0.000	0.000	0.0
	-10.000	+10.000	-10.0

With 5 VDC applied to pin 34 and Logic "1" applied to pin 12 the output at pin 28 shall increase to 5 ± 0.1 VDC in 32, 36 + 0.10 sec.

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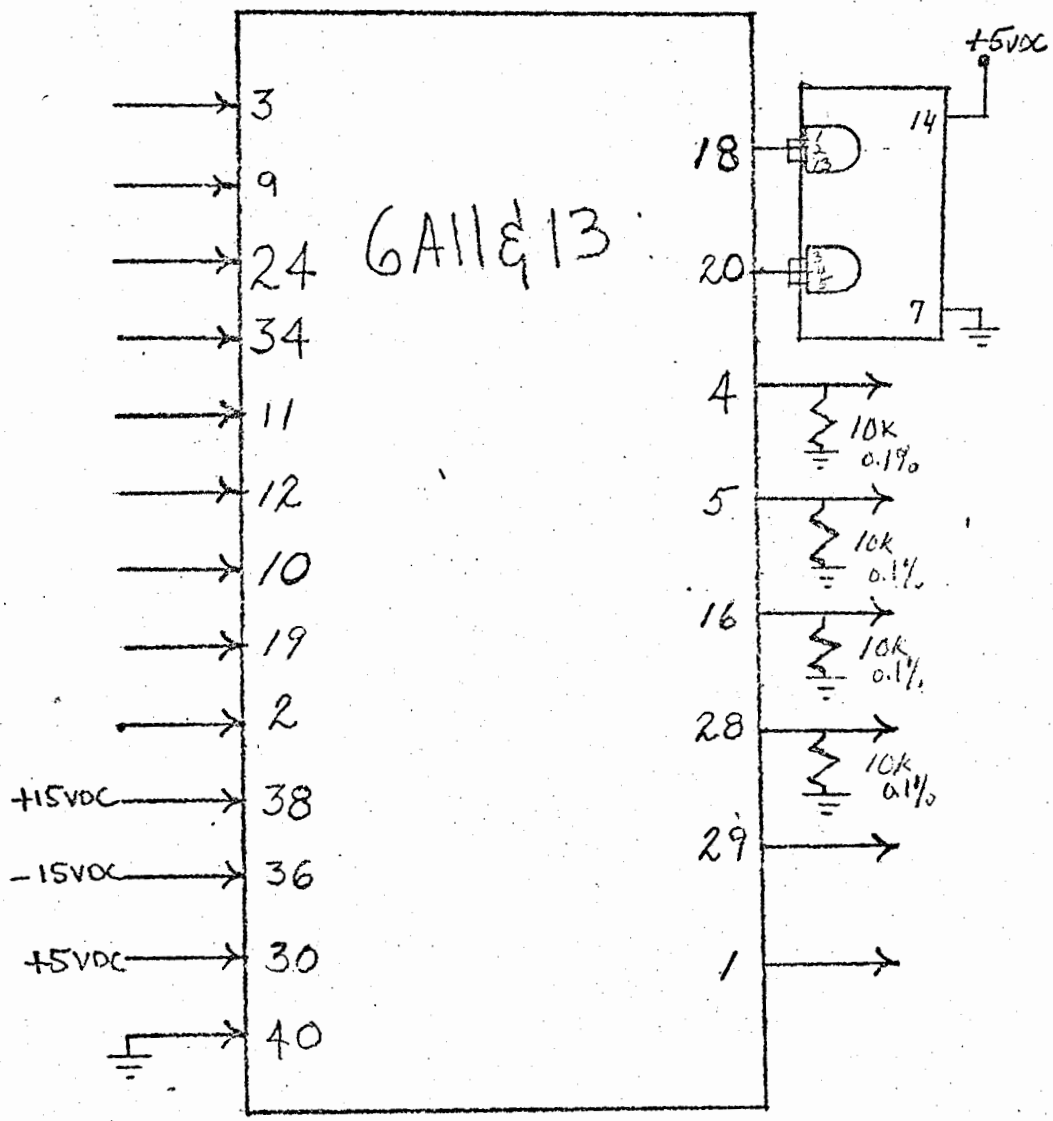


FIGURE I

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5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging and marking shall be in accordance with the requirements of Packaging Data Sheet (PDS), AMC Form 1029, 11746258.

6. NOTES

6.1 The notes specified in the general specification MIL-C-48411 are applied to this specification.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1240-A606

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